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Inventor(s)	Ding; Shunfeng

Center pole fixing structure

Abstract

Disclosed is a center pole fixing structure, including a center pole and a base. The base is provided with a center hole. The center pole is inserted into the center hole to be fixed. An anti-disengagement fixing structure is formed between the center pole and the base, achieves fixing by relative rotation of the center pole and the base, and constrains movement of the center pole in an axial direction thereof. A stopper for constraining the rotation of the center pole is disposed between the center pole and the base. The stopper is a movable pin, and the movable pin is controlled to enter or exit, thereby locking or unlocking the relative rotation of the center pole and the base. According to the disclosure, the center pole and the base are convenient to mount and dismount, and the fixed structure is firm and stable.

Inventors:	Ding; Shunfeng (Fujian, CN)
Applicant:	Aftercloud Technology Co., Ltd (Fujian, CN)
Family ID:	1000008778852
Assignee:	Aftercloud Technology Co., Ltd (Fujian, CN)
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Primary Examiner: McNichols; Eret C

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Chinese application serial no. 202223209996.6, filed on Dec. 1, 2022, Chinese application serial no. 202320275047.3, filed on December Feb. 21, 2023, and Chinese application serial no. 202222206211.3, filed on Aug. 22, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The present disclosure relates to the field of outdoor umbrellas, and particularly relates to a center pole fixing structure.

Description of Related Art

(3) An outdoor umbrella is widely used in leisure places such as plazas, beaches, parks and courtyards as an outdoor leisure utensil, and provides people with a comfortable leisure and shading space. The outdoor umbrella may be not only effective for sun protection and sunshade, but also is more conducive to constant promotion due to its inexpensive outdoor commercial advertising use.

(4) In general, the outdoor umbrella is composed of a collapsible umbrella body, a center pole and a base. The umbrella body is mounted on a top of the center pole and a bottom of the center pole is fixed on the base, wherein the center pole is generally square or round. In order to fix the center pole to the base, the existing outdoor umbrella is mostly fixed in a manner that one or more bolt pins pass through a center hole in the base to be locked with a screw/nut provided at the bottom of the center pole, or the center pole is directly inserted. The base is positioned using a bolt pin.

(5) However, on the one hand, since the outdoor environment is complicated, and the outdoor umbrella is required to bear a large self-weight and variable loads caused by a use operation or other processes in addition to loads such as wind and rain, the center pole is prone to rotation,

affecting the stability of the umbrella body. A gap is formed between the center pole and the base easily after long-term use, which results in a poor fixing effect between the center pole and the base, and produces wobbling, thereby causing safety hazards. Whereas, the wobbling of the center pole easily damages the thread of a bolt and even causes the deformation of each connecting structural part so as to often result in that the center pole cannot be disassembled normally. On the other hand, the base generally has a large weight, and massive stones serve as most of bases at present. Such an assembly is very inconvenient in operation, and wastes time and labor, and it is often required to lift the base to loosen the bolt during disassembly, thereby wasting time and labor. (6) Meanwhile, the out-of-roundness of an outer diameter of the center pole cannot be avoided during the production and manufacturing process, which results in decreased fitness degree of the center pole and the base to cause wobbling, further deteriorating the stability of the center pole. How to reinforce the firmness of the connection between the center pole and the base is the problem mainly to be solved by the present disclosure.

SUMMARY

(7) The present disclosure aims to solve one of the technical problems in the above-mentioned technology at least to some extent. To this end, an objective of the present disclosure is to provide a center pole fixing structure, which enhances the firmness and stability of the connection between the center pole and the base while improving the disassembly efficiency of a center pole.

(8) To achieve the above objective, an embodiment of the present disclosure provides a center pole fixing structure, including a center pole and a base, wherein the base is provided with a center hole, and the center pole is inserted into the center hole to be fixed. The center pole fixing structure is characterized in that, a matched anti-disengagement fixing structure is formed between the center pole and the base, achieves fixing by relative rotation of the center pole and the base, and constrains movement of the center pole in an axial direction thereof; a stopper for constraining the rotation of the center pole is disposed between the center pole and the base; and the stopper is a movable pin, and the movable pin is controlled to enter or exit, thereby locking or unlocking the relative rotation of the center pole and the base.

(9) According to the center pole fixing structure in the embodiment of the present disclosure, a center pole of an outdoor umbrella is mounted and dismounted quickly and meanwhile, the center pole may be prevented from disengagement by using the matched anti-disengagement fixing structure disposed between the center pole and the base. The mounting stability and safety of the center pole are improved by constraining the rotation of the center pole by providing the stopper between the center pole and the base.

(10) The center pole is constrained or unlocked by matching a hole or slot with the movable pin, thereby simplifying the manufacturing process and saving costs. The firmness of the center pole is further enhanced by the engagement of a flange ring on the outside of the center pole and a first flange or a support ring in the center hole, which prevents the center pole and the base from relative rotation to avoid disengagement, and well plays the roles of centering and wobbling resistance particularly when the fitness degree of the center pole and the base is reduced due to out-of-roundness of an outer diameter of the center pole or machining deformation.

(11) By providing the transition jacket in the base, the centering and windproof effects of a tube body of the center pole is better by disposing a transition sleeve in the base. In addition, the same base may also satisfy center poles having different outer diameters by replacing transition sleeves having different inner diameters, thereby saving costs.

(12) The present disclosure also allows for quick mounting and dismounting of the center pole. At the time of mounting, a downward pressing force is applied to the center pole after the center pole is inserted into the center hole to implement the connection of the anti-disengagement fixing structure. Meanwhile, the center pole extrudes an elastic locking pin, and then the center pole is rotated to clamp and lock the anti-disengagement fixing structure. The center pole is rotated until the elastic locking pin pops up to mount the center pole to the base. At the time of dismounting, the

elastic locking pin is pulled outwards so that the elastic locking pin is withdrawn from a pin hole, and the center pole is rotated to unlock the anti-disengagement fixing structure, thereby separating the center pole from the base. The operation is simple and fast.

(13) In addition, the center pole fixing structure provided according to the above embodiment of the present disclosure may also have the following additional technical features.

(14) Optionally, the anti-disengagement fixing structure is an anti-disengagement toothed ring and an anti-disengagement locking tab. The anti-disengagement toothed ring is provided with an anti-disengagement catch having a shape structure adaptive to a shape structure of the anti-disengagement locking tab.

(15) Further, when an anti-disengagement locking tab is provided on the base, when the anti-disengagement locking tab is disposed on the base, the anti-disengagement toothed ring is disposed at an end of the center pole inserted into the center hole, and the anti-disengagement toothed ring is integrally formed or detachably connected with the center pole.

(16) Further, the anti-disengagement toothed ring is integrally formed or detachably connected with the base when disposed on the base. The anti-disengagement locking tab is disposed in an end surface or an inner cavity of a tube body of the center pole.

(17) Optionally, the movable pin is an elastic telescopic pin, and the elastic telescopic pin is disposed on the base, and extends into a mating surface of the center pole and the base after the center pole and the base are in a rotation locking state to prevent the center pole and the base from rotation.

(18) Further, an anti-movement device is cooperatively disposed between contact surfaces of the anti-disengagement toothed ring and the anti-disengagement locking tab.

(19) Still further, the anti-movement device is a wedge or a bump or a ridge disposed between the contact surfaces.

(20) Further, a mounting block is disposed in the center hole of the base, and is provided with the stopper, and the stopper is a hole or a slot. The anti-disengagement toothed ring or the anti-disengagement locking tab is disposed at the center of the mounting block.

(21) Still further, a transition sleeve is embedded in the mounting block, and the transition sleeve is provided with a stopper mounting slot and a stop buckle. At the same time, the mounting block is provided with a stop hole or slot into which the stop buckle is inserted to prevent rotation of the transition sleeve.

(22) Still further, a bottom plate of the mounting block is provided with an anti-wobbling ring.

(23) Still further, a spacer is disposed between the mounting block and the base.

(24) Optionally, a flange ring is fixed at the outside of the center pole. A first flange is locked and fixed at the center hole in the base by a plurality of screws, and an inner diameter of the first flange cooperates with an outer diameter of the flange ring of the center pole.

(25) Further, a support ring is disposed under the first flange, and an end surface of the support ring cooperates with an end surface of the flange ring.

(26) Still further, a compensatory backing ring is disposed under the support ring, and the compensatory backing ring employs an elastic material.

(27) Optionally, the hole or slot is formed in an end of the center pole connected to the base.

(28) Optionally, the movable pin is a telescopic pin, and the base is provided with a sliding slot in which the movable pin slides.

(29) Optionally, the center pole is provided with a first anti-disengagement structure that cooperates with a second anti-disengagement structure on the base to form the anti-disengagement fixing structure.

(30) Further, the first anti-disengagement structure is a hook or a thread, and the second anti-disengagement structure has a shape structure adaptive to a shape structure of the first anti-disengagement structure.

- (31) Further, the movable pin is an eccentric movable pin assembly.
(32) Further, the elastic telescopic pin is an eccentric elastic telescopic pin assembly.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded view of an overall structure according to an embodiment of the present disclosure.
- (2) FIG. 2 is a partial section view showing the dismounting of a center pole according to an embodiment of the present disclosure.
- (3) FIG. 3 is an exploded view of an overall structure according to another embodiment of the present disclosure.
- (4) FIG. 4 is a partial section view showing the dismounting of a center pole according to another embodiment of the present disclosure.
- (5) FIG. 5 is an exploded view of an overall structure according to yet another embodiment of the present disclosure.
- (6) FIG. 6 is a partial section view showing the dismounting of a center pole according to yet another embodiment of the present disclosure.
- (7) FIG. 7 is an exploded view of an overall structure according to still another embodiment of the present disclosure.
- (8) FIG. 8 is a partial section view showing the dismounting of a center pole according to still another embodiment of the present disclosure.
- (9) FIG. 9 is a schematic diagram of mounting a center pole to a base according to an embodiment of the present disclosure.
- (10) FIG. 10 is a schematic diagram showing the dismounting of FIG. 9.
- (11) FIG. 11 is a schematic structural diagram of a bottom of a center pole according to different embodiments of the present disclosure.
- (12) FIG. 12 is an exploded view according to an embodiment of the present disclosure.
- (13) FIG. 13 is a structural diagram of a center pole according to an embodiment of the present disclosure.
- (14) FIG. 14 is a schematic structural diagram of a mounting block according to an embodiment of the present disclosure.
- (15) FIG. 15 shows a top view and a section view of an embodiment of FIG. 14.
- (16) FIG. 16 is an exploded view of an embodiment of FIG. 14.
- (17) FIG. 17 is a structural diagram of a center pole according to another embodiment of the present disclosure.
- (18) FIG. 18 shows a top view and a section view of a mounting base according to an embodiment of the present disclosure.
- (19) FIG. 19 is a schematic diagram showing the mounting of a center pole according to an embodiment of FIG. 12.
- (20) FIG. 20 is a partially enlarged view of a contact surface of an anti-disengagement fixing structure according to an embodiment of the present disclosure.
- (21) FIG. 21 is a schematic structural diagram of an anti-wobbling ring according to an embodiment.
- (22) FIG. 22 is a schematic structural diagram of another anti-wobbling ring according to an embodiment.
- (23) FIG. 23 is a schematic structural diagram of an anti-wobbling ring according to another embodiment.
- (24) FIG. 24 is a schematic structure showing a locking state and an unlocking state of an anti-

disengagement fixing structure according to the present disclosure.

(25) FIG. 25 is an exploded view of an embodiment in which a hook is used to serve as an anti-disengagement structure according to the present disclosure.

(26) FIG. 26 is a schematic diagram of mounting a center pole of FIG. 25 to a base.

(27) FIG. 27 is a schematic structural diagram showing that an eccentric elastic telescopic pin is used as a stopper according to the present disclosure.

(28) FIG. 28 is a diagram showing a locking process in FIG. 27.

DESCRIPTION OF THE EMBODIMENTS

(29) Embodiments of the present disclosure are described in detail below, and examples of the embodiments are shown in the drawings, wherein the same or similar numerals throughout represent same or similar elements or elements having the same or similar functions. The embodiments described below with reference to the drawings are exemplary, and are intended to be used for explaining the present disclosure, and are not to be understood as limiting the present disclosure.

(30) According to the present disclosure, a center pole of an outdoor umbrella is mounted and dismounted quickly and meanwhile, the center pole may be prevented from disengagement by using a matched anti-disengagement fixing structure disposed between the center pole and a base. The mounting stability and safety of the center pole are improved by constraining the rotation of the center pole by providing a stopper between the center pole and the base.

(31) The center pole is constrained or unlocked by matching a hole or slot with the movable pin, thereby simplifying the manufacturing process and saving costs. The firmness of the center pole is further enhanced by the engagement of a flange ring on the outside of the center pole and a first flange or a support ring in the center hole, which prevents the center pole and the base from relative rotation to avoid disengagement, and well plays the roles of centering and wobbling resistance particularly when the fitness degree of the center pole and the base is reduced due to out-of-roundness of an outer diameter of the center pole or machining deformation.

(32) By providing the transition jacket in the base, the centering and windproof effects of a tube body of the center pole is better by disposing a transition sleeve in the base. In addition, the same base may also satisfy center poles having different outer diameters by replacing transition sleeves having different inner diameters, thereby saving costs.

(33) The present disclosure also allows for quick mounting and dismounting of the center pole. At the time of mounting, a downward pressing force is applied to the center pole after the center pole is inserted into the center hole to implement the connection of the anti-disengagement fixing structure. Meanwhile, the center pole extrudes an elastic locking pin, and then the center pole is rotated to clamp and lock the anti-disengagement fixing structure. The center pole is rotated until the elastic locking pin pops up to mount the center pole to the base. At the time of dismounting, the elastic locking pin is pulled outwards so that the elastic locking pin is withdrawn from a pin hole, and the center pole is rotated to unlock the anti-disengagement fixing structure, thereby separating the center pole from the base. The operation is simple and fast.

(34) In order to better understand the above technical solutions, exemplary embodiments of the present disclosure will be described in more detail below with reference to the drawings. Although the exemplary embodiments of the present disclosure are illustrated in the drawings, it should be understood that the present disclosure may be implemented in various forms and should not be construed as being limited to the embodiments set forth herein. On the contrary, these embodiments are provided such that the present disclosure will be more thoroughly understood, and the scope of the present disclosure may be fully conveyed to those skilled in the art.

(35) In order to better understand the above technical solutions, the above technical solutions will be described in detail in conjunction with the drawings and specific embodiments of the present disclosure.

(36) According to an embodiment of the present disclosure as shown in FIG. 1 to FIG. 2, a center

pole fixing structure includes a center pole **1** and a base **9**, wherein the center pole **1** is a steel tube body, and the base **9** is provided with a center hole **91**. A matched anti-disengagement fixing structure is disposed between the center pole **1** and the base **9**, achieves fixing by relative rotation of the center pole **1** and the base **9**, and constrains movement of the center pole **1** in an axial direction thereof. A stopper for constraining the rotation of the center pole is disposed between the center pole **1** and the base **9**.

(37) The anti-disengagement fixing structure includes an anti-disengagement toothed ring **5** and an anti-disengagement locking tab **1712**. The anti-disengagement toothed ring **5** is provided with an anti-disengagement catches **51** (see FIGS. **1**, **3**, **6**, **7**, **11**, **13**, **18** and **24**) having a shape structure adaptive to a shape structure of the anti-disengagement locking tab **1712**. The stopper may be an elastic locking pin or a detachable movable bolt pin.

(38) As shown in FIG. **1** or FIG. **11**, a boss having a diameter smaller than an inner diameter of the center pole **1** is disposed in the center hole **91** of the base. The anti-disengagement toothed ring **5**, when fixed on the boss by a plurality of screws, forms a clamping slot **15** (as shown in FIG. **2**) for engaging with the anti-disengagement locking tab **1712** (the anti-disengagement locking tab **1712** may be provided in various manner in different embodiments, as shown in FIGS. **5** and **11**). The anti-disengagement toothed ring **5** may also be integrally formed with the base. The boss, the anti-disengagement toothed ring **5** and the base may be detachable from each other or integrally formed with each other. When the boss, the anti-disengagement toothed ring and the base are detachable from each other, the boss, the anti-disengagement toothed ring and the base are fixed with each other by screw locking or welding.

(39) A maximum diameter of the anti-disengagement toothed ring **5** is smaller than an inner diameter of the steel tube body of the center pole **1**. The anti-disengagement toothed ring **5** is provided with one or more anti-disengagement catches **51** that cooperate with corresponding structures at the bottom of the steel tube body to act as the anti-disengagement function of a steel tube. The quantity of the anti-disengagement catches **51** is four as shown in FIG. **3**, which may be adjusted according to the actual needs. The shape of the anti-disengagement catches **51** is not limited to that shown in the figures, as long as the anti-disengagement catches **51** may be cooperated and fixed with the base by relative rotation.

(40) An anti-disengagement locking tab **1712** is disposed at an end of the center pole **1** inserted into the center hole **91**, is arranged in an end surface or an inner cavity at a lower end of the steel tube body by welding or other forming processes, and has a hollow center, and an inner side edge is provided with at least one anti-disengagement toothed plate that cooperates with the anti-disengagement toothed ring **5** on the base to act as an anti-disengagement function for the steel tube. The anti-disengagement toothed ring **5** is cooperatively connected to the anti-disengagement locking tab **1712** to fix the center pole **1** to the base **9**. When the center pole **1** is in a locking state, the anti-disengagement locking tab **1712** is located under the anti-disengagement toothed ring **5**. The anti-disengagement locking tab **1712** and the anti-disengagement toothed ring constrain each other to prevent the center pole **1** from disengagement in the locking state as shown in (a) of FIG. **24**.

(41) A pin hole **131** is formed in an end of the center pole **1** connected to the base **9**. The pin hole **131** may be formed in a tube wall of the center pole **1** or on an accessory surrounding the outside of the center pole **1**. The base **9** is provided with a sliding slot **10**. A transversely-inserted anti-rotation bolt pin **7** is disposed in the sliding slot **10** and cooperates with the pin hole **131** to play the role of preventing the center pole **1** from rotation. When the center pole **1** is connected to the base **9**, the anti-rotation bolt pin **7** is clamped through the pin hole **131** as shown in (a) of FIG. **2**, thereby restraining the rotation of the center pole **1** and preventing the center pole **1** from being disengaged from the base **9** due to rotation or turning. When the anti-rotation bolt pin **7** enters the pin hole **131**, the center pole **1** is locked against rotation, and the center pole **1** is in a locking state with the base **9** at the moment. When the anti-rotation bolt pin **7** is withdrawn from the pin hole **131**, as shown in

(b) of FIG. 2, the center pole **1** may be rotated and is in an unlocking state at the moment. Preferably, a front end of the anti-rotation bolt pin **7** is provided with a rounded corner.

(42) An anti-rotation block **6** is disposed in the sliding slot **10** in the base, is used in cooperation with the anti-rotation bolt pin **7** and a compression spring **8**, and is used for controlling the anti-rotation bolt pin **7**. The anti-rotation bolt pin **7** may be toggled by the anti-rotation block **6** to lock or unlock the center pole **1**, so that the anti-rotation block **6** may be a separate operation element cooperating with the anti-rotation bolt pin **7**, or may be formed by an extension section integrally formed or fixedly connected with the anti-rotation bolt pin **7** to prevent rotation of the steel tube body. The compression spring **8** keeps the anti-rotation bolt pin **7** to be always inserted in the pin hole **131** of the steel tube body.

(43) A flange ring **13** is disposed at an outer side of one end of the center pole **1** connected to the base **9**, and an outer circumference of the flange ring cooperates with an inner hole of the first flange **21** to play the roles of centering and wobbling stop. An end surface of the flange ring **13** cooperates with an end surface of a support ring **3** to play the role of stopping wobbling. The flange ring **13** also plays the role of reducing the requirement on the accuracy of the dimension of the outer diameter of the center pole **1**. The center pole **1** may still have a stable and firm cooperation with the base **9** by means of the flange ring **13** when the dimension of the outer diameter is not uniform sufficiently.

(44) The first flange **21** is locked and fixed in the center hole in the base by a plurality of screws, and an inner circumference of the first flange **21** cooperates with the outer circumference of the flange ring **13** of the center pole **1** to alleviate the wobbling of the steel tube.

(45) A support ring **3** and a compensatory backing ring **4** are disposed under the first flange **21**. The support ring **3** is disposed on the compensatory backing ring **4** and is in contact with a lower end surface of the flange ring **13** of the center pole **1**. If there is no support ring **3** and the compensatory backing ring **4** is in direct contact with the flange ring **13** of the center pole **1**, the center pole **1** suffers from a relatively high resistance when in rotation. The support ring **3** is added to avoid that the flange ring **13** of the center pole **1** is in direct contact with the compensatory backing ring **4**. The compensatory backing ring **4** has certain elasticity, such that the lower end surface of the flange ring **13** of the steel tube body is more closely fitted to the support ring **3**.

(46) As shown in FIG. 9, the center pole **1** is inserted into the center hole **91** in the base **9** by aligning slots of the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** during mounting. The center pole **1** is pressed down such that the anti-disengagement locking tab **1712** is completely disposed under the anti-disengagement toothed ring **5** of the base. In the meantime, since the front end of the anti-rotation bolt pin **7** is provided with the rounded corner, the center pole **1** extrudes the anti-rotation bolt pin **7** as shown in (a) of FIG. 9 while being pressed down, such that the anti-rotation bolt pin **7** is completely withdrawn from the sliding slot in the base. The center pole **1** is then rotated until the anti-rotation bolt pin **7** pops out and is clamped into the pin hole **131** in the center pole **1**, that is, the tube mounting operation is completed. At this time, the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** are clamped together, so that the center pole **1** may neither rotate nor disengage from the base **9**. The above operation process may be performed with only one hand, saving time and labor.

(47) As shown in FIG. 10, when the center pole **1** needs to be dismantled, the anti-rotation block **6** is toggled outwards as shown in (a) of FIG. 10, and the anti-rotation bolt pin **7** is withdrawn from the pin hole **131** at this time. Then, the center pole **1** is rotated reversely, so that the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** are in an unlocking state as shown in (b) of FIG. 3, and the center pole **1** is removed, thereby completing the tube dismantling operation.

(48) In some embodiments, as shown in FIGS. 3 and 4, when the base **9** is thinner, a second flange **22** may be added in the center hole **91** to ensure a sufficient support height such that the center pole **1** is more stable after being fixed. It will be appreciated that, based on the same purpose, thickness

of the flange may be increased, or more flanges may be added, in order to obtain a sufficient support height. More specifically, the second flange **22** is located under the first flange **21** and has the same diameter as the first flange **21**. At this time, the locking state and the unlocking state of the center pole **1** and the base **9** are as shown in (a) to (b) of FIGS. **8**, respectively.

(49) In some embodiments, as shown in FIGS. **7** and **8**, the anti-rotation bolt pin **7** is in a form of a detachable movable locking pin **16** that cooperates with a pin hole **131** in the center pole **1** to limit the center pole **1**. The detachable movable locking pin **16** is removed so that the center pole **1** can be rotated relative to the base **9**. The pin hole **131** may be formed in the flange ring **13** of the center pole **1** or in the steel tube at a tail end of the center pole **1**. A detachable movable locking pin **16** may be placed in the sliding slot **10** in the base.

(50) In some embodiments, as shown in FIGS. **5** and **6**, the anti-disengagement toothed ring is interchanged with the anti-disengagement locking tab **1712** in positions. That is, the anti-disengagement toothed ring **5** is disposed at one end of the center pole **1** inserted into the center hole **91**, and the anti-disengagement locking tab **1712** is fixed at the center hole **91** in the base **9**. The anti-disengagement toothed ring **5** is placed in the inner cavity at the lower end of the center pole **1** by welding or other forming processes, and the inner side edge is provided with one or more anti-disengagement toothed plates, which structurally cooperate with the anti-disengagement locking tab **1712** on the base to act as an anti-disengagement function of the steel tube.

(51) The stopper matched is disposed between the contact surfaces of the anti-disengagement fixing structure, which reduces the requirement on the accuracy for the machining of the anti-disengagement fixing structure. For example, an edge of the anti-disengagement catch **51** or the anti-disengagement locking tab **1712** is a wedge portion. Alternatively, a surface of the anti-disengagement locking tab **1712** in contact with the anti-disengagement toothed ring **5** is provided with a bump or a ridge (not shown in the figures). The wedge portion or the bump may be disposed on the anti-disengagement locking tab **1712** or on the anti-disengagement toothed ring. When the anti-disengagement catch **51** and the anti-disengagement locking tab **1712** are locked together, a gap is present between the contact surfaces of the anti-disengagement catch **51** and the anti-disengagement locking tab **1712** to cause untight clamping, resulting in wobbling of the center pole **1**. The gap between the contact surfaces becomes smaller as the center pole **1** is rotated by the design of the wedge portion or the bump, thereby further improving the stability of the center pole **1**.

(52) In some embodiments, as shown in FIGS. **12-24**, the end of the center pole **1** inserted into the center hole **91** is provided with the anti-disengagement toothed ring **5**. A mounting block **17** is disposed in the center hole **91**. The mounting block **17** includes a sleeve **171**, and a bottom plate **1710** is welded on the sleeve **171**, may be detachably connected, or may be integrally formed with the sleeve **171**. The bottom plate **1710** is provided with the anti-disengagement locking tab **1712** that is cooperatively connected to the anti-disengagement toothed ring **5** to fix the center pole **1** to the base **9**. A locking pin is disposed in the mounting block **17**, and the flange ring **13** is fixed on the outside the tube body of the center pole **1**, and is provided with a pin hole **131**. The locking pin is clamped through the pin hole **131** when the center pole **1** is connected to the base **9**, thereby constraining the rotation of the center pole **1** and preventing the center pole **1** from disengagement from the base **9** due to rotation or turning.

(53) The locking pin is an elastic locking pin **173** or a press type locking pin or a telescopic elastic locking pin or a detachable movable locking pin **16**.

(54) The bottom plate **1710** of the mounting block **17** is provided with a threaded hole **1711** and a positioning notch **1713**. The threaded hole **1711** is used to tighten and fix the mounting block **17** in the center hole **91** in the base **9** by the cooperation of a locking disc **19** and screws **20**. One end of the elastic locking pin **173** passes through the positioning notch **1713** and is pivotally connected to the bottom plate **1710**.

(55) As shown in FIGS. **12-16**, the base **9** is provided with the center hole **91**. The center pole **1** is a

steel tube, and the flange ring **13** is welded on the outer side of the lower end of the tube body of the center pole, and the flange ring **13** is provided with the pin hole **131** that cooperates with the elastic locking pin **173** to play the role of preventing rotation. The outer circumference of the flange ring **13** cooperates with the mounting block **17** to play the roles of centering and wobbling stop. A clamping slot **15** having a diameter smaller than the inner diameter of the tube body is welded in the bottom of the center pole **1**. The anti-disengagement toothed ring **5** is welded on the clamping slot **15**, and has the maximum diameter smaller than the inner diameter of the tube body of the center pole **1**. The anti-disengagement toothed ring **5** is provided with one or more anti-disengagement catches **51** (not fully shown in the figures), which cooperate with the bottom plate **1710** of the mounting block **17** to act as an anti-disengagement function of the tube body. The quantity of the anti-disengagement catches **51** is four as shown in FIG. **13**, which may be adjusted according to the actual needs. The shape of the anti-disengagement catches **51** is not limited to that shown in the figures, as long as the anti-disengagement catches **51** may be cooperated and fixed with the base by relative rotation.

(56) The mounting block **17** includes a sleeve **171**, a transition sleeve **172**, and an elastic locking pin **173**. The sleeve **171** is welded to the bottom plate **1710**, and the bottom plate **1710** is provided with the anti-disengagement locking tab **1712** that cooperates with the anti-disengagement toothed ring **5** to prevent the center pole **1** from disengagement from the base **9**. The bottom plate **1710** is additionally provided with a positioning notch **1713**, a plurality of stop notches **1714** (not fully shown in the figure) and threaded holes **1711** (not fully shown in the figure). The mounting block **17** is tightened and fixed in the center hole **91** in the base **9** by the cooperation of the locking disc **19**, the screws **20** and the threaded holes **1711** in the bottom plate **1710**. A spring **1733** is mounted at a lower end of the elastic locking pin **173**, such that the elastic locking pin **173** always pops up.

(57) The transition sleeve **172** is embedded in the mounting block **17**, such that the tube body of the center pole **1** has good centering and wobbling stop effects. The transition sleeve **172** includes a stop buckle **1722**. The stop buckle **1722** is inserted into a stop notch **1714** in the bottom plate **1710** to prevent rotation of the transition sleeve **172**. The transition sleeve **172** is also provided with a stopper mounting slot **1721** that allows the elastic locking pin **173** to pass through the transition sleeve **17** to be fixed on the bottom plate **1710**. The same base **9** may satisfy center holes **1** having different outer diameters by replacing transition sleeves **172** having different inner diameters.

(58) The lower end of the elastic locking pin **173** is provided with a through hole **1732** such that the elastic locking pin protrudes out from the positioning notch **1713** in the bottom plate **1710** and is fixed by a transverse bolt pin. The upper end of the elastic locking pin **173** is provided with a bulge **1731**. The bulge **1731** cooperates with the pin hole **131** of the flange ring **13** on the outside of the center pole **1** to prevent rotation. A spring **1733** is also mounted at the lower end of the elastic locking pin **173**, such that the elastic locking pin **173** always pops up.

(59) The center pole **1** is inserted into the mounting block **17** of the center hole **91** in the base **9** by aligning slots of the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** during mounting. The tube body of the center pole **1** is pressed down such that the anti-disengagement toothed ring **5** is completely disposed under the bottom plate **1710** of the mounting block **17**. At the same time, the flange ring **13** will also press down the elastic locking pin **173**, and the tube body is rotated until the elastic locking pin **173** pops up to be clamped into the pin hole **131** of the flange ring **13**. The above operation process may be performed with only one hand, saving time and labor.

(60) When the center pole **1** is in a fixed state, the anti-disengagement toothed ring **5** is located under the anti-disengagement locking tab **1712** for the bottom plate **1710**, and the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** constrain each other to prevent the center pole **1** from disengagement, as shown in the locking state in (a) of FIG. **24**.

(61) When the center pole **1** needs to be disengaged, the bulge **1731** of the elastic locking pin **173** is pressed, so that the elastic locking pin **173** is withdrawn from the pin hole **131**. Then, the center

pole **1** may be rotated to withdraw the anti-disengagement toothed ring **5** at the lower end from the anti-disengagement locking tab **1712** for the bottom plate **1710** in an unlocking state as shown in (b) of FIG. **24**. Subsequently, the center pole **1** is pulled out.

(62) A spacer **18** is disposed between the mounting block **17** and the base **9**, and is preferably made of a material capable of slight deformation. The spacer **18** is added to avoid direct contact between the mounting block **17** and the surface of the base **9** so as to prevent scratching of the surface of the outer appearance of the base **9**, to reduce the relative sliding possibility of the mounting block **17** to some extent and to act as an anti-loosening effect on the other hand.

(63) The center pole **1** and the base **9** may also be in connection and cooperation by means of a hook.

(64) As shown in FIG. **20**, the stopper matched is disposed between the contact surfaces of the anti-disengagement fixing structure, which reduces the requirement on the accuracy for the machining of the anti-disengagement fixing structure. For example, the edge of the anti-disengagement catch **51** or the anti-disengagement locking tab **1712** is a wedge portion **176**. Alternatively, the surface of the anti-disengagement locking tab **1712** in contact with the anti-disengagement toothed ring **5** is provided with a bump **175** or a ridge (not shown in the figures). The wedge **176** or bump **175** may be provided on the anti-disengagement locking tab or on the anti-disengagement toothed ring. When the anti-disengagement catch **51** and the anti-disengagement locking tab **1712** are locked together, a gap is present between the contact surfaces of the anti-disengagement catch **51** and the anti-disengagement locking tab **1712** to cause untight clamping, resulting in wobbling of the center pole **1**. The gap between the contact surfaces becomes smaller as the center pole **1** is rotated by the design of the wedge portion **176** or the bump **175**, thereby further improving the stability of the center pole **1**.

(65) As shown in FIG. **21**-FIG. **22**, an anti-wobbling ring **177** is additionally disposed on the outside of the bottom plate **1710** of the mounting block **17**. When the anti-disengagement toothed ring **5** enters the center hole **91** in the base **9**, the outer circumference of the anti-disengagement toothed ring **5** cooperates with an inner wall of the anti-wobbling ring **177** to further improve the stability and firmness of the center pole **1** and the base **9**. (b) of FIG. **21** is a top view of the mounting block, wherein a black filled part refers to the anti-disengagement toothed and the center pole **1** in the locking state at this time. The anti-wobbling ring **177** may also be disposed the inside of the bottom plate **1710**, as shown in FIG. **22**.

(66) As shown in FIG. **17**-FIG. **18**, the difference from the previous embodiment is that the anti-disengagement toothed ring **5** is interchanged with the anti-disengagement locking tab **1712** in positions. At this time, the anti-disengagement toothed ring **5** is disposed at the center of the bottom plate **1710** of the mounting block **17**, and is fixed to the bottom plate **1710** through the clamping slot **15**. Similarly, the anti-disengagement toothed ring **5** is provided with a plurality of anti-disengagement catches **51**. The anti-disengagement locking tab **1712** is welded in the inner cavity at the lower end of the tube body of the center pole **1**, has a hollow center, and the inner edge is provided with one or more toothed plates, which cooperate with the anti-disengagement toothed ring **5** for the bottom plate **1710** of the mounting block **17** to act as an anti-disengagement function.

(67) As shown in FIG. **23**, the anti-wobbling ring **177** is disposed on the inside or outside of the anti-disengagement locking tab **1712** in the center pole **1**. When the center pole **1** is connected to the base **9**, the outer circumference of the anti-disengagement toothed ring **5** cooperates with the inner wall of the anti-wobbling ring **177** to further improve the stability and firmness of the center pole **1** and the base **9**. (d) of FIG. **23** is a top view of the mounting block, wherein a black filled portion refers to the anti-disengagement toothed ring.

(68) FIG. **19** shows a mounting process and a dismounting process of the center pole **1**, wherein operation steps during mounting are the same as the operation steps described in the above embodiment. First, the anti-disengagement toothed ring **5** of the center pole **1** is aligned with the anti-disengagement locking tab **1712** on the bottom plate **1710** of the mounting block **17**, and the

center pole **1** is inserted into the mount **17** of the hole **91** in the base **9**. The tube body of the center pole is pressed down, and at the same time, the flange ring **13** may also press down the elastic locking pin **173**. The tube body of the center pole **1** is rotated until the elastic locking pin **173** pops up to be clamped into the pin hole **131** in the flange ring **13**, thereby completing the tube mounting operation. At this time, the anti-disengagement toothed ring **5** and the anti-disengagement locking tab **1712** are clamped together, so that the center pole **1** may neither rotate nor disengage from the base **9**.

(69) When the center pole **1** needs to be dismantled, only the elastic locking pin **173** is pressed down to withdraw from the pin hole **131** in the flange ring **13**. At this time, the tube body of the center pole **1** is reversely rotated so that the anti-disengagement toothed ring **5** and the anti-disengagement plate **1712** are in the unlocking state, and the center pole **1** is removed, thereby completing the tube dismantling operation.

(70) In some embodiments, as shown in FIG. 25-FIG. 26, the center pole fixing structure includes: a base **9** provided with a center hole **91**; and a center pole **1** being a steel tube. A first anti-disengagement structure is disposed at an end of the center pole **1** inserted into the center hole **91**. As shown in FIG. 25, the first anti-disengagement structure is a hook **211**. A second anti-disengagement structure is disposed in the center hole **91** in the base **9**. The second anti-disengagement structure is a fixing slot **212** adaptive to a hook **211** in terms of a shape structure. The fixing slot **212** and the hook **211** are connected to fix the center pole **1** to the base **9** by the cooperation of the fixing slot **212** and the hook **211** at the lower end of the center pole **1**. The hook **211** may be welded on the center pole **1**, may be integrally formed with the center pole **1**, or may be detachably connected to the center pole **1**.

(71) A movable pin is disposed at the end of the center pole inserted into the center hole **91**. In one of embodiments, the movable pin is an eccentric movable pin assembly **213** as shown in FIG. 25. The base **9** is correspondingly provided with an eccentric movable pin slot **214** for insertion of the eccentric movable pin assembly **213** after rotation of the center pole **1** to achieve a locking function.

(72) In some embodiments, as shown in FIGS. 27-28, an elastic telescopic pin is an eccentric elastic telescopic pin assembly **22**, including an eccentric elastic telescopic member **221** and a spring **222**. The eccentric elastic telescopic member **221** protrudes towards a side of the anti-disengagement toothed ring **5** to form a fastener **2211** cooperating with the anti-disengagement toothed opening **52**. The anti-disengagement catch **51** is pressed down to extrude the elastic telescopic member **221** after the anti-disengagement toothed ring **5** is inserted into the mounting block **17**, as shown in (a) of FIG. 28. (b) of FIG. 28 shows that the anti-disengagement catch **51** further passes through the mounting block **17**. (c) of FIG. 28 shows that the anti-disengagement catch **51** is rotated, and the anti-disengagement toothed opening **52** has approached the fastener **2211**. (d) of FIG. 28 shows that the eccentric elastic telescopic member **221** is clamped into the anti-disengagement toothed opening **52** due to pushing of the spring **222**, thereby completing the locking. In the case of unlocking, the eccentric elastic telescopic member **221** is pressed to disengage the fastener **2211** from the anti-disengagement toothed opening, and then the above steps are performed reversely to achieve unlocking.

(73) Of course, one or more anti-disengagement toothed openings **52** may be provided.

(74) In the description of the present disclosure, it needs to be understood that, the terms “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “counterclockwise” and the like indicate an orientation or positional relationship based on the orientation or positional relationship shown in the drawings, and are only for facilitating the description of the present disclosure and simplifying the description, rather than indicating or implying that the device or element indicated must have a particular orientation, be constructed and operate in a particular orientation. Therefore, the terms should not be construed as limiting the

present disclosure.

(75) In the present disclosure, unless definitely specified and defined otherwise, a first feature being “above” or “below” a second feature may include that the first feature and the second feature are in direct contact or that the first feature and the second feature are not in direct contact but are in contact through another feature therebetween. Moreover, a first feature being “above”, “over”, and “on” a second feature includes that the first feature is right above and obliquely above the second feature, or only indicates that the first feature is at a higher level than the second feature. A first feature being “under”, “below” and “beneath” a second feature includes that the first feature is right below and obliquely below the second feature, or only indicates that the first feature has a smaller height than the second feature.

(76) In the description in this specification, the description of reference terms “one embodiment”, “some embodiments”, “an example”, “a specific example”, or “some examples” or the like means that particular features, structures, materials, or characteristics described in combination with the embodiment or example are included in at least one embodiment or example of the present disclosure. In this specification, illustrative description of the above terms should not be understood as necessarily referring to the same embodiment or example. Moreover, the specific features, structures, materials, or characteristics described may be combined in any one or more embodiments or examples in an appropriate manner. Furthermore, different embodiments or examples described in this specification may be joined and combined by those skilled in the art.

(77) Although the embodiments of the present disclosure have been shown and described above, it will be appreciated that the above embodiments are illustrative and not to be construed as limiting the present disclosure. Those of ordinary skill in the art may make changes, modifications, substitutions, and variations to the above embodiments within the scope of the present disclosure.

Claims

1. A center pole fixing structure, comprising a center pole and a base, the base being provided with a center hole, and the center pole being inserted into the center hole to be fixed; wherein an anti-disengagement fixing structure is formed between the center pole and the base, achieves fixing by relative rotation of the center pole and the base, and constrains movement of the center pole in an axial direction thereof; a stopper for constraining the rotation of the center pole is disposed between the center pole and the base; and the stopper is a movable pin, and the movable pin is controlled to enter or exit, thereby locking or unlocking the relative rotation of the center pole and the base; and a flange ring is fixed at the outside of the center pole; a first flange is locked and fixed at the center hole in the base by a plurality of screws, and an inner diameter of the first flange cooperates with an outer diameter of the flange ring of the center pole.
2. The center pole fixing structure according to claim 1, wherein the anti-disengagement fixing structure is an anti-disengagement toothed ring and an anti-disengagement locking tab; the anti-disengagement toothed ring is provided with an anti-disengagement catch having a shape structure adaptive to a shape structure of the anti-disengagement locking tab.
3. The center pole fixing structure according to claim 2, wherein when the anti-disengagement locking tab is disposed on the base, the anti-disengagement toothed ring is disposed at an end of the center pole inserted into the center hole, and the anti-disengagement toothed ring is integrally formed or detachably connected with the center pole.
4. The center pole fixing structure according to claim 2, wherein the anti-disengagement toothed ring is integrally formed or detachably connected with the base when disposed on the base; and the anti-disengagement locking tab is disposed in an end surface or an inner cavity of a tube body of the center pole.
5. The center pole fixing structure according to claim 2, wherein an anti-movement device is cooperatively disposed between contact surfaces of the anti-disengagement toothed ring and the

anti-disengagement locking tab.

6. The center pole fixing arrangement according to claim 5, wherein the anti-movement device is a wedge or a bump or a ridge disposed between the contact surfaces.
 7. The center pole fixing structure according to claim 2, wherein a mounting block is disposed in the center hole of the base, and is provided with the stopper; the stopper is a hole or a slot; and the anti-disengagement toothed ring or the anti-disengagement locking tab is disposed at the center of the mounting block.
 8. The center pole fixing structure according to claim 7, wherein a transition sleeve is embedded in the mounting block, and the transition sleeve is provided with a stopper mounting slot and a stop buckle; and at the same time, the mounting block is provided with a stop hole or slot into which the stop buckle is inserted to prevent rotation of the transition sleeve.
 9. The center pole fixing structure according to claim 7, wherein a bottom plate of the mounting block is provided with an anti-wobbling ring.
 10. The center pole fixing structure according to claim 7, wherein a spacer is disposed between the mounting block and the base.
 11. The center pole fixing structure according to claim 7, wherein the hole or slot is formed in an end of the center pole connected to the base.
 12. The center pole fixing structure according to claim 1, wherein the movable pin is an elastic telescopic pin, and the elastic telescopic pin is disposed on the base, and extends into a mating surface of the center pole and the base after the center pole and the base are in a rotation locking state to prevent the center pole and the base from rotation.
 13. The center pole fixing structure according to claim 12, wherein the elastic telescopic pin is an eccentric elastic telescopic pin assembly.
 14. The center pole fixing structure according to claim 1, wherein a support ring is disposed under the first flange, and an end surface of the support ring cooperates with an end surface of the flange ring.
 15. The center pole fixing structure according to claim 14, wherein a compensatory backing ring is disposed under the support ring, and the compensatory backing ring employs an elastic material.
 16. The center pole fixing structure according to claim 1, wherein the movable pin is a telescopic pin, and the base is provided with a sliding slot in which the movable pin slides.
 17. The center pole fixing structure according to claim 1, wherein the center pole is provided with a first anti-disengagement structure that cooperates with a second anti-disengagement structure on the base to form the anti-disengagement fixing structure.
 18. The center pole fixing structure according to claim 17, wherein the first anti-disengagement structure is a hook or a thread, and the second anti-disengagement structure has a shape structure adaptive to a shape structure of the first anti-disengagement structure.
 19. The center pole fixing structure according to claim 17, wherein the movable pin is an eccentric movable pin assembly.
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